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Cost-effectiveness Analysis of Various Strategies in the Diagnostic Management of Pulmonary Embolism

Matthijs Oudkerk, MD, PhD; Edwin J. R. van Beek, MD; Wim L. J. van Putten, MSc; Harry R. Büller, MD, PhD

Background: Since the clinical diagnosis of pulmonary embolism is unreliable, various objective diagnostic methods (or combinations thereof) are advocated. Pulmonary angiography is the accepted reference method but is considered less suitable for initial screening due to its invasive nature. Therefore, at least nine different diagnostic management strategies employing invasive and noninvasive diagnostic tests are used in clinical practice. We assessed the cost-effectiveness of these strategies to help identify the optimal approach.

Methods: Based on assumptions derived from published data, we calculated mortality, morbidity, adequacy for the indication of anticoagulant therapy, and associated diagnostic and therapeutic costs using a decision analytic model. Additionally, a cost-effectiveness analysis was performed using incremental costs per additional life saved.

Results: The analysis identified three separate categories

of diagnostic management strategies: (1) Treatment of all patients (or of those with an abnormal perfusion scan) results in the lowest mortality and morbidity rates but the highest costs due to inappropriate treatment of 55% to 70% of patients. (2) Pulmonary angiography strategies with or without prior perfusion-ventilation lung scintigraphy and ultrasonography of the legs have comparable low mortality and morbidity rates, costs savings of approximately 40%, and inappropriate treatment in fewer than 5% of patients. (3) Use of perfusion-ventilation scintigraphy with or without ultrasonography results in unacceptably high mortality rates.

Conclusions: At present, the optimal diagnostic management strategy should include pulmonary angiography. Use of perfusion-ventilation lung scintigraphy and ultrasonography results in a 40% to 50% reduction in the number of patients requiring pulmonary angiography and is cost-effective.

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TRADITIONALLY, THE diagnosis of pulmonary embolism was made on the basis of clinical history and physical examination, usually with the aid of blood gas analysis, chest radiography, and electrocardiography.¹⁻⁴ Although it was acknowledged that these diagnostic methods were nonspecific, the vast majority of patients were treated with anticoagulant agents.⁵ With the evaluation of pulmonary angiography in prospective studies, it became apparent that the diagnosis of pulmonary embolism could be confirmed in only 30% of patients with suspected pulmonary embolism.⁶⁻¹⁰ Hence, institution and continuation of anticoagulant therapy on the basis of clinical diagnosis alone would be unjustified in approximately 70% of patients.

Notwithstanding the general acceptance of pulmonary angiography as the reference method for the diagnosis of pulmonary embolism, it was considered less suitable as a screening method due to its invasive nature and limited availability and the morbidity related to the procedure.⁹⁻¹³ This led to the search for less invasive techniques, and, after its introduction in 1964, the perfusion lung scan was evaluated as a screening test for pulmonary embolism.¹⁴⁻¹⁶ Various studies have shown that a normal perfu-

See Methods on next page

METHODS

PREVALENCE OF VENOUS THROMBOEMBOLISM

To calculate outcomes of the various diagnostic management strategies, the following assumptions were made with respect to the prevalence of pulmonary embolism and the sensitivity and specificity of the individual diagnostic tests. Among patients with clinically suspected pulmonary embolism, angiography provides proof in approximately 30%,^{10,20,21,28-30} and the prevalence of coexisting deep vein thrombosis in the leg, as shown by contrast venography, is approximately 70%.^{12,31,35} The number of patients with suspected pulmonary embolism who only have deep vein thrombosis is considered to approximate zero and was set at 0.1%.³⁶

The outcome probabilities of the individual diagnostic test results in the presence or absence of venous thromboembolism in patients with suspected pulmonary embolism are summarized in **Table 1**.

PULMONARY ANGIOGRAPHY

Conventional pulmonary angiography is generally regarded as the reference method for the diagnosis of pulmonary embolism.^{10,20,37} Recent studies using long-term clinical outcome have revealed that the sensitivity and specificity are 98% and 96%, respectively.^{10,29,38,39} The mortality and morbidity (defined as severe allergic reaction, cardiac perforation, and rhythm disturbances requiring treatment) rates associated with this diagnostic modality were estimated to be 0.2% and 1.9%, respectively.⁹⁻¹¹ The costs of the procedure were calculated to be \$575 (500 European currency units [ECU]; 1 ECU = \$1.15 US). These costs and those for the other diagnostic procedures include the costs of material, personnel, and analysis and are based on a survey of 10 European and Canadian medical centers.

PERFUSION-VENTILATION LUNG SCINTIGRAPHY

A normal perfusion scan is considered to exclude clinically important pulmonary embolism.^{17,18} The chance of obtaining a normal perfusion scan in the presence or absence of pulmonary embolism is 0.2% or 20%, respectively.^{10,18} The chance of a high-probability lung scan re-

sult (defined as one or more segmental perfusion defects with normal ventilation) is 41% if pulmonary emboli are present and 2% if they are absent.^{10,12,20-26} Patients with pulmonary emboli have a chance of 42.8% of having subsegmental perfusion defects and a 16% chance of having matched segmental perfusion-ventilation defects (together classified as nonhigh-probability scan). In the absence of pulmonary emboli, these chances are 62% and 16%, respectively. It was assumed that perfusion-ventilation scintigraphy is not associated with morbidity or mortality. The costs were calculated to be \$115 for the perfusion scan and \$288 for the ventilation scan.

REAL-TIME ULTRASONOGRAPHY

Real-time ultrasonography is presently regarded as the most sensitive and specific noninvasive test for the detection of deep vein thrombosis of the legs. The sensitivity and specificity of this method for all thrombi in patients with clinically suspected pulmonary embolism were calculated to be 79% and 99%, respectively, by adapting available data from studies in patients with suspected deep vein thrombosis.^{40,41} Neither morbidity nor mortality was considered to be associated with this procedure, and the costs were valued at \$75.

NATURAL HISTORY AND TREATMENT

Untreated pulmonary embolism is associated with a mortality rate of 26% in the first 8 weeks after diagnosis, whereas this can be reduced to 2% with appropriate anticoagulant treatment.⁴²⁻⁴⁴ Over the same period, nonfatal recurrence of pulmonary embolism for untreated and treated pulmonary embolism was estimated to occur in 26% and 2% of the cases, respectively.^{28,35,42-45} Mortality from pulmonary embolism as a result of untreated and treated deep vein thrombosis was assumed to be 12% and 0.7%, respectively, over a 3-month period.⁴⁴⁻⁴⁷ Untreated and treated deep vein thromboses are associated with nonfatal pulmonary embolism in 15% and 2% of cases, respectively.⁴⁴⁻⁴⁷

Treatment for either pulmonary embolism or deep vein thrombosis was considered to include 7 days of hospitalization for intravenous heparin therapy, followed by oral anticoagulant therapy for a period of 3 months. The total treatment costs amounted to \$2490, including medication, laboratory tests, and medical supervision. Three months of

sion scan virtually excludes clinically important pulmonary embolism (and that anticoagulant therapy can be safely withheld from these patients),^{17,18} whereas an abnormal perfusion scan constitutes insufficient proof for the presence of pulmonary embolism.^{10,15,16,18} Subsequently, ventilation scanning was added to improve the diagnostic utility of the perfusion scan.¹⁹⁻²¹ A number of studies have since shown that a high-probability perfusion-ventilation scan, defined as one or more segmental or greater perfusion defects with normal ventilation, corresponds with angiographically proven

pulmonary embolism in approximately 90% of patients.^{10,12,20,21} In spite of numerous attempts to classify the remaining perfusion-ventilation scan results, none have been shown to be clinically useful for therapeutic management.²²⁻²⁵ Therefore, it has been suggested that all remaining perfusion-ventilation scan results should be classified as nondiagnostic or nonhigh probability.^{26,27} This test result is found in about half of all patients with a clinically suspected pulmonary embolism, whereas only 20% to 30% of these patients actually have pulmonary embolism.^{10,20,21,28-30} To avoid pulmonary angiogra-

anticoagulant treatment was considered to be associated with a mortality rate of 0.25% and a morbidity rate of 1%, both due to major hemorrhage.⁴⁸⁻⁵²

DIAGNOSTIC MANAGEMENT STRATEGIES

We defined nine different diagnostic management strategies that are presently used in clinical practice. These strategies were (1) treatment of all patients with suspected pulmonary embolism without the performance of any objective diagnostic procedures; (2) perfusion scan and treatment of all patients with an abnormal scan; (3) perfusion scan, with pulmonary angiography if an abnormal scan is found, and treatment of all patients with angiographically proven pulmonary embolism; (4) perfusion-ventilation scan and treatment of all patients with a high-probability scan; (5) perfusion-ventilation scan, with pulmonary angiography if a nonhigh-probability lung scan is found, and treatment of all patients with either a high-probability scan or angiographically proven pulmonary embolism; (6) pulmonary angiography in all patients and treatment if pulmonary embolism is shown; (7) real-time B-mode ultrasonography of the legs, with pulmonary angiography in all patients with a normal ultrasound result, and treatment of all patients with either an abnormal ultrasound result or angiographically proven pulmonary embolism; (8) perfusion-ventilation scan, with ultrasonography in case of a nonhigh-probability scan result, and treatment in all patients with either a high-probability scan result or an abnormal ultrasound result; and (9) perfusion-ventilation scan with ultrasonography if a nonhigh-probability lung scan result is found, followed by pulmonary angiography if ultrasonography is normal, and treatment of all patients with a high-probability scan result, an abnormal ultrasonography result, or angiographically proven pulmonary embolism. In addition, as a reference, we added a "clairvoyant" diagnostic strategy in which all patients with pulmonary embolism were correctly identified (without the use of a diagnostic test) and treated, whereas those without pulmonary embolism did not receive anticoagulant therapy.

For strategies including perfusion-ventilation scintigraphy, it was assumed that the ventilation scan was only performed if segmental or larger perfusion defects were shown. Other perfusion scan abnormalities were defined as non-high probability.

phy, different additional diagnostic strategies, such as the noninvasive detection of deep vein thrombosis in the legs and blood tests to ascertain degradation of fibrin clots, are advocated for patients with nonhigh-probability lung scan results.^{28,31-34}

Thus, there appear to be both controversy and confusion with regard to the optimal diagnostic strategy that should be used for the detection of pulmonary embolism in symptomatic patients. In particular, there exists uncertainty about the appropriate role and timing of pulmonary angiography, lung scanning, and tests for deep

ANALYSIS

The outcomes of the various diagnostic strategies were calculated and checked with the Decision Maker 3.0 computer model.⁵³ The primary classification of the strategies was according to the overall mortality rate. Furthermore, the morbidity rate and the percentage of patients who inappropriately received no treatment as well as the percentage of patients who received anticoagulant therapy unnecessarily were determined. Finally, the costs of the diagnostic workup, the treatment, and the combination of both were assessed. In addition, a cost-effectiveness analysis was performed. The average costs per life gained, compared with no treatment given, were calculated. The primary classification of the strategies was according to the mortality per 1000 investigated symptomatic patients. The marginal cost-effectiveness ratio (incremental costs per extra life saved) was used to further evaluate the use of noninvasive tests prior to pulmonary angiography. The costs associated with complications of the diagnostic procedures or the therapeutic management thereof were not included in the analysis. Furthermore, an additional cost calculation was made with assumptions adjusted for the health care system of the United States to identify possible changes in the order of preferred strategies as a result of varying costs, since the costs of these services may be substantially higher than those in Europe and Canada. Costs were based on a survey of three representative medical centers in the United States and were estimated to be \$500 for perfusion lung scintigraphy, \$450 for ventilation lung scanning, \$250 for compression B-mode ultrasonography, \$2500 for pulmonary angiography, and \$6000 for total treatment costs.

SENSITIVITY ANALYSIS

A number of sensitivity analyses were performed, with total mortality as an end point, including prevalence of pulmonary embolism (10% to 50%), sensitivity of pulmonary angiography (96% to 100%), specificity of pulmonary angiography (92% to 100%), mortality from pulmonary angiography (0.1% to 0.4%), mortality from untreated pulmonary embolism (10% to 40%), mortality from anticoagulant therapy (0.15% to 0.35%), and fatal recurrence rate of pulmonary embolism in adequately treated patients (1% to 15%). The effects of these variations on overall costs were also assessed.

vein thrombosis in the diagnostic workup of these patients. The outcome of any diagnostic strategy in pulmonary embolism will influence treatment decisions and consequently will influence mortality and morbidity rates and costs associated with the chosen diagnostic-therapeutic approach. We therefore undertook a cost-effectiveness analysis for a number of diagnostic management strategies that are used in current clinical practice. After review of the literature, assumptions were derived and used in a model to calculate the effects of these strategies on mortality, morbidity, diagnostic efficacy, and costs.

RESULTS

The diagnostic management strategies can be subclassified by mortality rates into three categories compared with the reference strategy (**Table 2**). The first category encompasses two strategies with either perfusion scan only or no diagnostic tests at all (strategies 2 and 1, category I). The second category comprises five strategies that utilize various noninvasive methods for the detection of venous thromboembolism, always including pulmonary angiography (strategies 9, 7, 5, 3, and 6; category II). The last category comprises strategies that use perfusion-ventilation lung scanning with or without a noninvasive test for deep vein thrombosis but that do not include pulmonary angiography (strategies 8 and 4, category III).

REFERENCE DIAGNOSTIC MANAGEMENT STRATEGY

This "clairvoyant" diagnostic management strategy assumes that, without the use of any diagnostic method, only those patients who have pulmonary embolism receive anticoagulant therapy. The mortality and morbidity rates for this strategy are 0.68% and 0.9%, respectively, while the average overall costs per patient are \$750 (**Table 2**).

CATEGORY I

These two strategies, in which either all symptomatic patients or only those with an abnormal perfusion scan receive anticoagulant therapy, have the lowest mortality and morbidity rates of the three categories (mortality rates, 0.83% and 0.85%, respectively, for strategies 2 and 1). However, these strategies result in the unnecessary treatment of 55% to 70% of the initial cohort of patients with suspected pulmonary embolism. The overall costs for these management strategies are approximately \$2300, which is two to four times more expensive than the overall costs of the other two categories (**Table 2**).

CATEGORY II

The five strategies in this category, in which pulmonary angiography is used with or without noninvasive diagnostic tests for venous thromboembolism, are associated with a mortality rate of approximately 1% and a morbidity rate of 2% to 3% (strategies 9, 7, 5, 3, and 6). The five diagnostic management strategies in this category are associated with a 20-fold reduction of the overtreatment of patients (2.24% to 4.10%) compared with the strategies in category I. The percentages of patients who require pulmonary angiography are 62%, 82%, 72%, 86%, and 100% for strategies 9, 7, 5, 3, and 6, respectively. The overall costs vary between \$1377 and \$1455 (**Table 2**).

Of the five strategies, strategy 9 (lung scanning and

ultrasonography prior to pulmonary angiography) has the lowest mortality, morbidity, and number of patients in whom anticoagulant treatment is inappropriately withheld. The overall costs of this approach are comparable to those of the other four strategies in this category.

CATEGORY III

The two diagnostic management strategies in this category, which do not use pulmonary angiography but rely on lung scanning with or without ultrasonography, result in the highest mortality and morbidity rates (mortality rates, 2.56% and 4.89%, respectively, for strategies 8 and 4). Furthermore, both strategies induce inappropriate withholding of anticoagulant therapy in 8% to 18% of patients with suspected pulmonary embolism, which is 10 to 70 times higher than rates associated with the strategies in category II. The overall costs for each of the strategies approach those of the "clairvoyant" strategy (**Table 2**).

COST-EFFECTIVENESS ANALYSIS

The costs per life gained were calculated for each of the strategies compared with a strategy in which none of the patients with symptomatic pulmonary embolism received anticoagulant therapy. The mortality per 1000 symptomatic patients for this approach was calculated to be 78.1 (**Table 3**). The results of the cost-effectiveness analysis indicate that the same three categories could be distinguished, as was found for the classification of diagnostic management strategies.

The two strategies with only perfusion lung scanning or no diagnostic workup at all and treatment of (virtually) all patients (category I) have the lowest mortality rates; however, the costs per life gained are in the order of \$34 000. The five strategies of category II, which employ pulmonary angiography with or without noninva-

Table 1. Outcome Probabilities of Diagnostic Test Results in the Presence or Absence of Venous Thromboembolism in Patients With Suspected Pulmonary Embolism

	Absent, %	Present, %
Pulmonary embolism	70	30
Perfusion-ventilation scan result		
Normal	20	0.2
Subsegmental defects	62	43
Segmental matched defects	16	16
Segmental mismatched defects	2	41
Pulmonary angiography result		
Normal	96	2
Abnormal	4	98
Deep vein thrombosis	79	21
Ultrasound scan result		
Normal	99	21
Abnormal	1	79

sive tests for venous thromboembolism, have comparable cost-effectiveness (costs per life gained vary between \$20 000 and \$20 985), whereas the mortality is approximately nine per 1000 investigated patients. The reduction in costs per life gained for these strategies compared with those in category I is approximately 40%. The last category, which encompasses two strategies with only noninvasive tests for venous thromboembolism, is associated with high mortality rates, whereas the costs per life gained are the lowest (approximately \$17 000).

In **Table 4**, the incremental costs per life gained with the use of ultrasonography or lung scanning or a combination thereof prior to pulmonary angiography are shown in order of decreasing cost-effectiveness. By performing ultrasonography in all patients prior to angiog-

raphy, the extra cost involved to save one extra life is \$958, whereas the cost with the combination of perfusion-ventilation scanning and ultrasonography is \$46 000. The incremental costs per life saved of obtaining only a perfusion scan before angiography are twice as high as using the more extensive strategy of perfusion-ventilation scanning followed by ultrasonography.

SENSITIVITY AND OTHER ANALYSES

We performed multiple sensitivity analyses varying the prevalence of pulmonary embolism, sensitivity and specificity of pulmonary angiography, mortality from pulmonary angiography, mortality from untreated pulmonary embolism, mortality from anticoagulant therapy, and fa-

Table 2. Diagnostic Management Strategies With Resulting Mortality, Morbidity, Diagnostic Efficacy, Costs, and Required Pulmonary Angiography Rate in Patients With Suspected Pulmonary Embolism (PE)

Strategy	Mortality Rate, %*	Morbidity Rate, %*	Inappropriately Received No Treatment, %	Inappropriate Treatment, %	Costs, \$†			Pulmonary Angiography Required, %
					Diagnostic	Therapeutic	Overall	
Correct diagnosis and treatment of all patients (reference)	0.68	0.90	0	0	0	750	750	0
2. Perfusion scan; treatment if scan is abnormal	0.83	1.48	0.1	55.9	115	2140	2255	0
1. No diagnostic test; treatment of all patients	0.85	1.60	0	69.9	0	2490	2490	0
9. Perfusion-ventilation scan, with ultrasonography if scan shows nonhigh probability and angiography if ultrasound is normal; treatment if scan shows high probability, if ultrasound is abnormal, or if angiography shows PE	0.87	2.17	0.2	4.1	610	845	1455	61.8
7. Ultrasonography in all patients, with angiography if ultrasound is normal; treatment if ultrasound is abnormal or if angiography shows PE	0.92	2.57	0.3	3.5	550	829	1379	82.5
5. Perfusion-ventilation scan, with angiography if scan shows nonhigh probability; treatment if scan shows high probability or if angiography shows PE	0.94	2.42	0.5	3.6	615	826	1441	72.2
3. Perfusion scan with angiography if scan is abnormal; treatment if PE is shown	1.02	2.72	0.8	2.2	610	787	1397	85.9
6. Angiography in all patients; treatment if PE is shown	1.04	2.98	0.7	2.8	575	802	1377	100
8. Perfusion-ventilation scan, with ultrasonography if scan shows nonhigh probability; treatment if scan shows high probability or if ultrasound is abnormal	2.56	2.74	7.9	1.9	254	600	854	0
4. Perfusion-ventilation scan; treatment only if scan shows high probability	4.89	5.00	17.8	1.4	200	342	542	0

*Associated with pulmonary angiography, anticoagulant treatment, or recurrent venous thromboembolism.

†Average costs per patient (converted from European currency units).

Table 3. Mortality and Cost-effectiveness in Terms of Life Gained in Patients With Suspected Pulmonary Embolism (PE)

Strategy	Mortality, per 1000 Patients	Lives Saved per 1000 Patients*	Costs per Extra Life Saved, \$†
2. Perfusion scan; treatment if scan is abnormal	8.3	69.8	32 302
1. No diagnostic test; treatment of all patients	8.5	69.6	35 772
9. Perfusion-ventilation scan, with ultrasonography if scan shows nonhigh probability and angiography if ultrasound is normal; treatment if scan shows high probability, if ultrasound is abnormal, or if angiography shows PE	8.7	69.4	20 968
7. Ultrasonography in all patients, with angiography if ultrasound is normal, treatment if ultrasound is abnormal or if angiography shows PE	9.2	68.9	20 000
5. Perfusion-ventilation scan, with angiography if scan shows nonhigh probability; treatment if scan shows high probability or if angiography shows PE	9.4	68.7	20 985
3. Perfusion scan, with angiography if scan is abnormal; treatment if PE is shown	10.2	67.9	20 552
6. Angiography in all patients; treatment if PE is shown	10.4	67.7	20 306
8. Perfusion-ventilation scan, with ultrasonography if scan shows nonhigh probability; treatment if scan shows high probability or if ultrasound is abnormal	25.6	52.5	16 313
4. Perfusion-ventilation scan; treatment only if scan shows high probability	48.9	29.2	18 544
No treatment	78.1	...	...

*Extra lives saved compared with reference (no treatment).

†Converted from European currency units.

tal recurrence rate of pulmonary embolism due to ineffectiveness of anticoagulant therapy (see the "Methods" section).

Table 5 shows the results of the sensitivity analyses, with end-point mortality, if only one parameter was varied. Strategy 9 proved to be the best strategy of categories II and III in all situations. Furthermore, this strategy had the lowest mortality of all nine evaluated strategies if the prevalence of pulmonary embolism became less than 10%, if the mortality due to anticoagulant therapy became higher than 0.32%, or if the mortality due to pulmonary angiography became less than 0.15%. Also, assuming a higher recurrence rate of pulmonary embolism during adequate anticoagulant therapy did not alter the order of the strategies. Finally, overall costs of the various strategies, as indicated in Table 2, were only influenced by variation of prevalence. With increasing prevalence of pulmonary embolism from 30% to 70%, the cost savings of strategy 9 compared with strategy 2 decreased from 40% to 5%.

With recalculations using costs for services in the United States, the cost savings obtained by using strategies from category II were approximately 30% compared with the strategies of category I. Pulmonary angiography alone was always more expensive than strategies 7, 9, and 5.

COMMENT

In this study, we considered the consequences of the use of nine different diagnostic management strategies for patients with clinically suspected pulmonary embolism that are presently used in medical practice. Our analysis assessed the overall mortality, morbidity, and costs of these strategies using published data for the diagnostic accuracy and complication rates of the various diagnostic techniques employed.

When considering which strategy should be selected for clinical use, the present analysis may provide two important guidelines. First, no or minimal use of objective diagnostic techniques (such as perfusion scanning only) for the detection or exclusion of pulmonary embolism results in unnecessary anticoagulant treatment in a large proportion of patients. Although this approach is associated with the lowest mortality and morbidity rates, it is not cost-effective (Table 3). Second, at present, an adequate diagnostic management strategy should include pulmonary angiography. Such a strategy is associated with mortality and morbidity rates that are comparable with the approach in which no or minimal objective diagnos-

Table 4. Increasing Incremental Costs per Extra Life Gained of Additional Noninvasive Tests for Venous Thromboembolism Compared With Pulmonary Angiography Alone

Strategy*	Extra Costs per Extra Life Saved, \$†
7. Ultrasonography in all patients, with angiography if ultrasound is normal; treatment if ultrasound is abnormal or if angiography shows PE	958
9. Perfusion-ventilation scan, with ultrasonography if scan shows nonhigh probability and angiography if ultrasound is normal; treatment if scan shows high probability, if ultrasound is abnormal, or if angiography shows PE	46 000
5. Perfusion-ventilation scan, with angiography if scan shows nonhigh probability; treatment if scan shows high probability or if angiography shows PE	65 550
3. Perfusion scan, with angiography if scan is abnormal; treatment if PE is shown	92 000

*PE indicates pulmonary embolism.

†Converted from European currency units.

tic methods are used. Strategies that employ pulmonary angiography as part of the diagnostic workup were highly cost-effective, with savings of approximately 40%. In contrast, the use of only noninvasive diagnostic methods (perfusion-ventilation lung scanning with or without real-time ultrasonography of the leg veins) results in the strategies with the lowest costs, but, due to the inappropriate withholding of anticoagulant therapy, these approaches have unacceptably high mortality rates. Varying the assumptions for the true prevalence of pulmonary embolism, mortality from untreated pulmonary embolism, mortality and morbidity rates for both anticoagulant therapy and pulmonary angiography, and recurrence rate of pulmonary embolism with adequate anticoagulant therapy (in a sensitivity analysis) did not affect these major conclusions from the analysis. Moreover, reanalysis for costs of services applicable to the United States revealed a similar order of preferred strategies.

Further evaluation of the five strategies that use pulmonary angiography revealed that the mortality and morbidity rates and costs differ marginally. Since it is generally accepted that noninvasive methods are initially preferred, the optimal timing of performing pulmonary angiography becomes clinically relevant. As our analysis suggests, the benefit of performing pulmonary angiography is greatest for patients in whom perfusion-ventilation lung scanning and, in case of a nonhigh-probability lung scan outcome, real-time ultrasonography of the leg veins have yielded nondiagnostic and normal test outcomes, respectively (Table 2, strategy 9). With such a strategy, the need for this invasive procedure can be avoided in approximately 40% of patients with suspected pulmonary embolism. Alternatively, if neither noninvasive method is available, pulmonary angiography should be performed in all patients.

Analysis of the incremental costs incurred to save one additional life showed that B-mode ultrasonography prior to pulmonary angiography is highly cost-effective, while the combination of perfusion-ventilation scanning followed by B-mode ultrasonography in addition to pulmonary angiography results in incremental costs of \$46 000 per extra life saved (Table 4). However, the strategy that only uses B-mode ultrasonography is associated with an extra need for invasive pulmonary angiography in 20% more patients than the preferred strategy 9 (Table 2), while the overall costs per patient do not differ. The calculated incremental costs per life saved of \$46 000 compare favorably with the costs of other commonly accepted medical diagnostic tests or treatment interventions.⁵⁴⁻⁵⁹ It should be realized that the present analysis focused on lives saved and did not include an assessment of quality of life, which may be affected by the use of invasive tests or their complications or by the long-term use of medication.

In conclusion, this cost-effectiveness analysis suggests that, at present, the most efficient approach appears to be a combination of perfusion-ventilation lung scan-

Table 5. Effects of the Variation of Selected Parameters on the Order of Mortality of Diagnostic Strategies for Pulmonary Embolism

Parameter	Range, %	Three Strategies With the Lowest Mortality*	
		Lowest Value	Highest Value
Prevalence	10-50	9†, 5, 7	9, 7, 5
Mortality due to anticoagulant therapy	0.15-0.35	9, 7, 5	9†, 7, 5
Mortality due to pulmonary angiography	0.1-0.4	9†, 7, 5	9, 7, 5
Sensitivity of pulmonary angiography	96-100	9, 7, 5	9†, 7, 5
Specificity of pulmonary angiography	92-100	9, 7, 5	9, 7, 5
Fatal recurrence rate of pulmonary embolism with adequate anticoagulant therapy	1-15	9, 7, 5	9†, 7, 5

*Excluding strategies 1 and 2. See the "Methods" section for definitions of the diagnostic management strategies.

†Strategy 9 was also better than strategies 1 and 2.

‡Strategy 9 was surpassed only by strategy 1.

ning, B-mode ultrasonography of the leg veins, and pulmonary angiography. The initial investigation should be perfusion lung scanning, since a normal perfusion scan result rules out clinically relevant pulmonary embolism. In case of segmental or greater perfusion defects, a ventilation scan should be obtained. If a high-probability test result is found, the diagnosis of pulmonary embolism becomes highly likely, and in most patients this will serve as the basis for long-term anticoagulant therapy. In patients with all other perfusion (ventilation) defects, a B-mode ultrasound test of the leg veins should be performed, and, if results are normal, a pulmonary angiogram is indicated.

Recently suggested alternatives to this approach, such as serial noninvasive techniques for deep vein thrombosis detection and plasma D-dimer assays in patients with nondiagnostic lung scan or B-mode ultrasound test results, are promising and possibly more cost-effective options,^{28,33} but they need further clinical evaluation before they can be addressed in a cost-effectiveness analysis.

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